

 INAF ISTITUTO NAZIONALE DI ASTRONOMIA OSSERVATORIO ASTRONOMICO DI ARCETRI	FATE MONTHLY REPORT Doc.No : FATE-001 Version : 1.0 Date : 03-07-2024	 CONSORZIO LaMMA
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FORECAST SYSTEM FOR ATMOSPHERE AND TURBULENCE at the VERY LARGE TELESCOPE

Performance assessment
Annex
June 2024



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1 NIGHT1: Contingency tables

Tables in section 1 are included in the range: 1-22.

wind speed ms^{-1}	Observations		
	ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	252	112
	4.2 < ws < 7.6	156	197
	ws > 7.6	14	95
Number of points 1943 ; PC=71.1%; EBD=1.5%; POD1=59.7%; POD2=48.8%; POD3=83.5%			

Table 1: night1 - Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed ms^{-1}	Observations		
	ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	397	23
	4.2 < ws < 7.6	25	367
	ws > 7.6	0	14
Number of points 1943 ; PC=94.9%; EBD=0.0%; POD1=94.1%; POD2=90.8%; POD3=96.6%			

Table 2: night1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H)

wind speed ms^{-1}	Observations		
	ws < 12.0	12.0 < ws < 18.0	ws > 18.0
Model data	ws < 12.0	1465	32
	12.0 < ws < 18.0	10	413
	ws > 18.0	0	8
Number of points 1943 ; PC=97.2%; EBD=0.0%; POD1=99.3%; POD2=91.2%; POD3=66.7%			

Table 3: night1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H) and thresholds (12.0,18.0)

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wind direction degree		North	East	South	Observations	West
Model data	North	1451	36	0		0
	East	16	46	7		0
	South	3	4	78		0
	West	30	2	0		0
Number of points 1673 ; PC=94.1%; EBD=0.3%; POD1=96.7%; POD2=52.3%; POD3=91.8%; POD4=NaN%						

Table 4: night1 - Contingency table for wind direction (degree) in standard configuration

wind direction degree		North	East	South	Observations	West
Model data	North	1494	13	0		0
	East	6	70	7		0
	South	0	5	78		0
	West	0	0	0		0
Number of points 1673 ; PC=98.1%; EBD=0.0%; POD1=99.6%; POD2=79.5%; POD3=91.8%; POD4=NaN%						

Table 5: night1 - Contingency table for wind direction (degree) processed with AR (1H)

relative humidity percent		rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	427	106	0
	7.1 < rh < 15.3	327	594	126
	rh > 15.3	2	79	282
Number of points 1943 ; PC=67.1%; EBD=0.1%; POD1=56.5%; POD2=76.3%; POD3=69.1%				

Table 6: night1 - Contingency table for relative humidity (percent) in standard configuration

relative humidity percent		rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	725	25	0
	7.1 < rh < 15.3	31	747	3
	rh > 15.3	0	7	405
Number of points 1943 ; PC=96.6%; EBD=0.0%; POD1=95.9%; POD2=95.9%; POD3=99.3%				

Table 7: night1 - Contingency table for relative humidity (percent) processed with AR (1H)

relative humidity percent		rh < 50.0	50.0 < rh < 70.0	rh > 70.0
Model data	rh < 50.0	1943	0	0
	50.0 < rh < 70.0	0	0	0
	rh > 70.0	0	0	0
Number of points 1943 ; PC=100.0%; EBD=0.0%; POD1=100.0%; POD2=NaN%; POD3=NaN%				

Table 8: night1 - Contingency table for relative humidity (percent) processed with AR (1H) and thresholds (50,70) percent

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	704	207	0
	1.9 < pwv < 3.5	99	396	75
	pwv > 3.5	0	0	392
Number of points 1873 ; PC=79.7%; EBD=0.0%; POD1=87.7%; POD2=65.7%; POD3=83.9%				

Table 9: night1 - Contingency table for precipitable water vapor (mm) in standard configuration

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	788	15	0
	1.9 < pwv < 3.5	15	587	4
	pwv > 3.5	0	1	463
Number of points 1873 ; PC=98.1%; EBD=0.0%; POD1=98.1%; POD2=97.3%; POD3=99.1%				

Table 10: night1 - Contingency table for precipitable water vapor (mm) processed with AR (1H)

FATE - Monthly Report

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
	see < 0.6	191	65	0
Model data	0.6 < see < 0.8	139	87	43
	see > 0.8	5	38	92
Number of points 660 ; PC=56.1%; EBD=0.8%; POD1=57.0%; POD2=45.8%; POD3=68.1%				

Table 11: night1 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
	see < 0.6	324	25	0
Model data	0.6 < see < 0.8	30	155	10
	see > 0.8	0	12	115
Number of points 660 ; PC=88.5%; EBD=0.0%; POD1=91.5%; POD2=80.7%; POD3=92.0%				

Table 12: night1 - Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
	see < 0.6	332	0	0
Model data	0.6 < see < 0.8	0	174	0
	see > 0.8	3	16	135
Number of points 660 ; PC=97.1%; EBD=0.5%; POD1=99.1%; POD2=91.6%; POD3=100.0%				

Table 13: night1 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
	see < 0.6	354	0	0
Model data	0.6 < see < 0.8	0	192	0
	see > 0.8	0	0	125
Number of points 660 ; PC=100.0%; EBD=0.0%; POD1=100.0%; POD2=100.0%; POD3=100.0%				

Table 14: night1 - Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.24

FATE - Monthly Report

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	33	29	22
	$3.4 < \tau < 5.8$	41	149	93
	$\tau > 5.8$	6	42	91
Number of points 506 ; PC=54.0%; EBD=5.5%; POD1=41.2%; POD2=67.7%; POD3=44.2%				

Table 15: night1 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	66	5	1
	$3.4 < \tau < 5.8$	14	192	27
	$\tau > 5.8$	0	23	178
Number of points 506 ; PC=86.2%; EBD=0.2%; POD1=82.5%; POD2=87.3%; POD3=86.4%				

Table 16: night1 - Contingency table for wavefront coherence time (ms) processed with AR (1H) and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	74	0	82
	$3.4 < \tau < 5.8$	0	215	82
	$\tau > 5.8$	6	5	124
Number of points 506 ; PC=81.6%; EBD=17.4%; POD1=92.5%; POD2=97.7%; POD3=43.1%				

Table 17: night1 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	80	0	1
	$3.4 < \tau < 5.8$	0	215	1
	$\tau > 5.8$	0	5	205
Number of points 506 ; PC=98.8%; EBD=0.2%; POD1=100.0%; POD2=97.7%; POD3=99.0%				

Table 18: night1 - Contingency table for wavefront coherence time (ms) processed with AR (1H) and accuracy 1.22

FATE - Monthly Report

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	60	79
	0.6 < glf < 0.7	41	59
	glf > 0.7	34	37
Number of points 508 ; PC=36.8%; EBD=11.2%; POD1=44.4%; POD2=33.7%; POD3=34.3%			

Table 19: night1 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	109	16
	0.6 < glf < 0.7	26	130
	glf > 0.7	0	29
Number of points 508 ; PC=81.7%; EBD=0.2%; POD1=80.7%; POD2=74.3%; POD3=88.9%			

Table 20: night1 - Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	98	12
	0.6 < glf < 0.7	3	159
	glf > 0.7	34	4
Number of points 508 ; PC=84.6%; EBD=11.6%; POD1=72.6%; POD2=90.9%; POD3=78.3%			

Table 21: night1 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	135	0
	0.6 < glf < 0.7	0	175
	glf > 0.7	0	0
Number of points 508 ; PC=99.8%; EBD=0.2%; POD1=100.0%; POD2=100.0%; POD3=99.0%			

Table 22: night1 - Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.14

2 DAY1: Contingency tables

Tables in section 2 are included in the range: 23-32.

wind speed ms^{-1}	Observations		
	ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	517	127
	4.6 < ws < 7.4	206	76
	ws > 7.4	7	35
Number of points 2010 ; PC=76.3%; EBD=0.6%; POD1=70.8%; POD2=31.9%; POD3=90.2%			

Table 23: day1 - Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed ms^{-1}	Observations		
	ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	684	31
	4.6 < ws < 7.4	46	182
	ws > 7.4	0	25
Number of points 2010 ; PC=94.2%; EBD=0.0%; POD1=93.7%; POD2=76.5%; POD3=98.6%			

Table 24: day1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H)

wind speed ms^{-1}	Observations		
	ws < 12.0	12.0 < ws < 18.0	ws > 18.0
Model data	ws < 12.0	1358	42
	12.0 < ws < 18.0	6	423
	ws > 18.0	0	8
Number of points 2010 ; PC=95.7%; EBD=0.0%; POD1=99.6%; POD2=89.4%; POD3=82.7%			

Table 25: day1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H) and thresholds (12.0,18.0)

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wind direction degree		North	East	South	Observations West
Model data	North	1221	21	0	4
	East	12	43	16	0
	South	7	5	48	18
	West	130	7	10	124
Number of points 1666 ; PC=86.2%; EBD=0.8%; POD1=89.1%; POD2=56.6%; POD3=64.9%; POD4=84.9%					

Table 26: day1 - Contingency table for wind direction (degree) in standard configuration

wind direction degree		North	East	South	Observations West
Model data	North	1352	2	0	5
	East	6	73	4	0
	South	0	1	67	0
	West	12	0	3	141
Number of points 1666 ; PC=98.0%; EBD=0.0%; POD1=98.7%; POD2=96.1%; POD3=90.5%; POD4=96.6%					

Table 27: day1 - Contingency table for wind direction (degree) processed with AR (1H)

relative humidity percent		Observations		
		rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	490	264	0
	8.9 < rh < 17.9	227	621	138
	rh > 17.9	0	50	220
Number of points 2010 ; PC=66.2%; EBD=0.0%; POD1=68.3%; POD2=66.4%; POD3=61.5%				

Table 28: day1 - Contingency table for relative humidity (percent) in standard configuration

relative humidity percent		Observations		
		rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	669	29	0
	8.9 < rh < 17.9	48	898	6
	rh > 17.9	0	8	352
Number of points 2010 ; PC=95.5%; EBD=0.0%; POD1=93.3%; POD2=96.0%; POD3=98.3%				

Table 29: day1 - Contingency table for relative humidity (percent) processed with AR (1H)

relative humidity percent		Observations		
		rh < 50.0	50.0 < rh < 70.0	rh > 70.0
Model data	rh < 50.0	2010	0	0
	50.0 < rh < 70.0	0	0	0
	rh > 70.0	0	0	0
Number of points 2010 ; PC=100.0%; EBD=0.0%; POD1=100.0%; POD2=NaN%; POD3=NaN%				

Table 30: day1 - Contingency table for relative humidity (percent) processed with AR (1H) and thresholds (50,70) percent

precipitable water vapor mm		Observations		
		pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	711	139	0
	2.1 < pwv < 3.7	126	465	182
	pwv > 3.7	0	26	361
Number of points 2010 ; PC=76.5%; EBD=0.0%; POD1=84.9%; POD2=73.8%; POD3=66.5%				

Table 31: day1 - Contingency table for precipitable water vapor (mm) in standard configuration

precipitable water vapor mm		Observations		
		pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	816	19	0
	2.1 < pwv < 3.7	21	602	8
	pwv > 3.7	0	9	535
Number of points 2010 ; PC=97.2%; EBD=0.0%; POD1=97.5%; POD2=95.6%; POD3=98.5%				

Table 32: day1 - Contingency table for precipitable water vapor (mm) processed with AR (1H)

3 NIGHT2: Contingency tables

Tables in section 3 are included in the range: 33-42.

wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	214	108	12
	4.2 < ws < 7.6	90	202	171
	ws > 7.6	61	84	934
Number of points 1876 ; PC=72.0%; EBD=3.9%; POD1=58.6%; POD2=51.3%; POD3=83.6%				

Table 33: night2 - Contingency table for wind speed (ms^{-1}) in standard configuration

wind direction degree		Observations			
		North	East	South	West
Model data	North	1419	27	7	0
	East	39	47	4	0
	South	10	14	57	0
	West	32	0	2	0

Number of points 1658 ; PC=91.9%; EBD=1.0%; POD1=94.6%; POD2=53.4%; POD3=81.4%; POD4=NaN%

Table 34: night2 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent		Observations		
		rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	306	112	0
	7.1 < rh < 15.3	412	567	89
	rh > 15.3	18	53	319

Number of points 1876 ; PC=63.5%; EBD=1.0%; POD1=41.6%; POD2=77.5%; POD3=78.2%

Table 35: night2 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	667	124	0
	1.9 < pwv < 3.5	136	411	166
	pwv > 3.5	0	1	301

Number of points 1806 ; PC=76.4%; EBD=0.0%; POD1=83.1%; POD2=76.7%; POD3=64.5%

Table 36: night2 - Contingency table for precipitable water vapor (mm) in standard configuration

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	218	72	0
	0.6 < see < 0.8	204	115	130
	see > 0.8	51	88	220
Number of points 1098 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 37: night2 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	431	0	0
	0.6 < see < 0.8	0	222	0
	see > 0.8	42	53	350
Number of points 1098 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 38: night2 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	291	176	56
	$3.4 < \tau < 5.8$	35	143	95
	$\tau > 5.8$	3	19	83
Number of points 901 ; PC=57.4%; EBD=6.5%; POD1=88.4%; POD2=42.3%; POD3=35.5%				

Table 39: night2 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	323	17	94
	$3.4 < \tau < 5.8$	3	316	94
	$\tau > 5.8$	3	5	140
Number of points 901 ; PC=86.5%; EBD=10.8%; POD1=98.2%; POD2=93.5%; POD3=42.7%				

Table 40: night2 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

ground layer fraction dimensionless		Observations		
		glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	109	88	40
	0.6 < glf < 0.7	125	202	160
	glf > 0.7	27	64	134
Number of points 949 ; PC=46.9%; EBD=7.1%; POD1=41.8%; POD2=57.1%; POD3=40.1%				

Table 41: night2 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless		Observations		
		glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	228	13	48
	0.6 < glf < 0.7	6	336	42
	glf > 0.7	27	5	286
Number of points 949 ; PC=89.6%; EBD=7.9%; POD1=87.4%; POD2=94.9%; POD3=76.1%				

Table 42: night2 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

4 DAY2: Contingency tables

Tables in section 4 are included in the range: 43-46.

wind speed ms^{-1}		Observations		
		ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	536	130	1
	4.6 < ws < 7.4	190	88	98
	ws > 7.4	4	20	943
Number of points 2010 ; PC=78.0%; EBD=0.2%; POD1=73.4%; POD2=37.0%; POD3=90.5%				

Table 43: day2 - Contingency table for wind speed (ms^{-1}) in standard configuration

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	1273	27	0
	East	0	28	2
	South	3	21	60
	West	94	0	12

Number of points 1666 ; PC=89.6%; EBD=0.2%; POD1=92.9%; POD2=36.8%; POD3=81.1%; POD4=89.7%

Table 44: day2 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	413	237
	8.9 < rh < 17.9	304	651
	rh > 17.9	0	47

Number of points 2010 ; PC=63.5%; EBD=0.8%; POD1=57.6%; POD2=69.6%; POD3=59.2%

Table 45: day2 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	692	138
	2.1 < pwv < 3.7	145	447
	pwv > 3.7	0	45

Number of points 2010 ; PC=73.8%; EBD=0.0%; POD1=82.7%; POD2=71.0%; POD3=63.5%

Table 46: day2 - Contingency table for precipitable water vapor (mm) in standard configuration

5 NIGHT3: Contingency tables

Tables in section 5 are included in the range: 47-56.

wind speed <i>ms</i> ⁻¹		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	189	107	6
	4.2 < ws < 7.6	136	124	106
	ws > 7.6	35	133	973
Number of points 1809 ; PC=71.1%; EBD=2.3%; POD1=52.5%; POD2=34.1%; POD3=89.7%				

Table 47: night3 - Contingency table for wind speed (*ms*⁻¹) in standard configuration

FATE - Monthly Report

wind direction degree		Observations			
		North	East	South	West
Model data	North	1352	36	6	0
	East	28	29	7	0
	South	30	23	57	0
	West	23	0	0	0

Number of points 1591 ; PC=90.4%; EBD=2.3%; POD1=94.3%; POD2=33.0%; POD3=81.4%; POD4=NaN%

Table 48: night3 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent		Observations		
		rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	250	148	0
	7.1 < rh < 15.3	390	535	132
	rh > 15.3	29	49	276

Number of points 1809 ; PC=58.7%; EBD=1.6%; POD1=37.4%; POD2=73.1%; POD3=67.6%

Table 49: night3 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	694	199	0
	1.9 < pwv < 3.5	109	297	254
	pwv > 3.5	0	0	213

Number of points 1766 ; PC=68.2%; EBD=0.0%; POD1=86.4%; POD2=59.9%; POD3=45.6%

Table 50: night3 - Contingency table for precipitable water vapor (mm) in standard configuration

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	220	49	9
	0.6 < see < 0.8	186	87	88
	see > 0.8	65	97	231
Number of points 1032 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 51: night3 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	410	0	0
	0.6 < see < 0.8	0	208	0
	see > 0.8	61	25	328
Number of points 1032 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 52: night3 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	230	69	5
	$3.4 < \tau < 5.8$	63	219	126
	$\tau > 5.8$	0	28	103
Number of points 843 ; PC=65.5%; EBD=0.6%; POD1=78.5%; POD2=69.3%; POD3=44.0%				

Table 53: night3 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	287	5	85
	$3.4 < \tau < 5.8$	6	306	85
	$\tau > 5.8$	0	5	149
Number of points 843 ; PC=88.0%; EBD=10.1%; POD1=98.0%; POD2=96.8%; POD3=46.7%				

Table 54: night3 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

ground layer fraction dimensionless		Observations		
		glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	47	57	32
	0.6 < glf < 0.7	177	209	147
	glf > 0.7	6	67	149
Number of points 891 ; PC=45.5%; EBD=4.3%; POD1=20.4%; POD2=62.8%; POD3=45.4%				

Table 55: night3 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless		Observations		
		glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	215	0	35
	0.6 < glf < 0.7	9	330	35
	glf > 0.7	6	3	293
Number of points 891 ; PC=94.1%; EBD=4.6%; POD1=93.5%; POD2=99.1%; POD3=80.7%				

Table 56: night3 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

6 DAY3: Contingency tables

Tables in section 6 are included in the range: 57-60.

wind speed ms^{-1}		Observations		
		ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	405	102	1
	4.6 < ws < 7.4	322	85	101
	ws > 7.4	3	51	940
Number of points 2010 ; PC=71.1%; EBD=0.2%; POD1=55.5%; POD2=35.7%; POD3=90.2%				

Table 57: day3 - Contingency table for wind speed (ms^{-1}) in standard configuration

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wind direction degree	Observations			
	North	East	South	West
Model data	North	1281	36	0
	East	0	32	9
	South	2	8	52
	West	87	0	13

Number of points 1666 ; PC=89.1%; EBD=0.1%; POD1=93.5%; POD2=42.1%; POD3=70.3%; POD4=82.2%

Table 58: day3 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	381	251
	8.9 < rh < 17.9	288	579
	rh > 17.9	48	105

Number of points 2010 ; PC=57.2%; EBD=2.4%; POD1=53.1%; POD2=61.9%; POD3=53.1%

Table 59: day3 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	717	172
	2.1 < pwv < 3.7	120	428
	pwv > 3.7	0	30

Number of points 2010 ; PC=73.7%; EBD=0.0%; POD1=85.7%; POD2=67.9%; POD3=61.9%

Table 60: day3 - Contingency table for precipitable water vapor (mm) in standard configuration